

Hardware Requirements Specification (HRS)

Project: Autonomous Solar-Powered LoRaWAN Soil Monitoring Node

Prepared by: AmudarIO Research LLC

1. Project Overview & Design Flexibility

The objective is to design a compact, ultra-low-power, solar-rechargeable LoRaWAN sensor node. The device will act as a bridge between an external 5V RS485 soil probe and a LoRaWAN gateway (ChirpStack). The hardware will be deployed in agricultural environments and must survive extreme seasonal temperatures while maintaining long-range connectivity.

- **Sample Reference:** The design should take inspiration from industrial soil sensor products (such as the Makerfabs AgroSense LoRaWAN Industrial Soil Sensor) in terms of power efficiency, outdoor reliability, and robust battery-powered operation.
- **Component & Architectural Flexibility:** The developer is free to propose alternative components, ICs, or topological improvements based on design experience, supply availability, or performance optimizations. Any significant structural or component substitutions must be submitted to and approved by AmudarIO Research LLC prior to final implementation.

2. Core Processing, RF & Positioning

- **Microcontroller:** ESP32-C6 (RISC-V 32-bit processor) selected for ultra-low deep sleep power consumption, advanced power management unit (PMU), integrated LP/RTC memory, and native ESP-IDF / Arduino support.
- **LoRa Radio:** SX1262-based LoRa module featuring an integrated LNA (Low Noise Amplifier) to maximize receive sensitivity for ground-level agricultural deployments.
- **LoRa Regional Setting:** RF matching and module selection must be optimized for the 868MHz ISM band (EU868 standard).
- **Antenna Interfaces:** Standard SMA female connectors for both LoRa and GPS antennas (no U.FL / IPEX).
- **GPS/GNSS Module:** Low-power GPS module with an SMA antenna interface. The power supply to the GPS module must be fully switchable (gateable via ESP32-C6 GPIO) to ensure zero current draw when not actively obtaining a fix.

- Memory/Storage: Non-Volatile Storage (NVS) via ESP32 flash and Low-Power (LP) / RTC memory for persistent retention of LoRaWAN OTAA session credentials and frame counters across reboots and deep sleep cycles.

3. Power Management & Solar Charging

- Primary Power: 3.7V Lithium-Ion battery (18650 or LiPo pouch).
- Solar Input: MPPT or optimized solar charge controller IC rated for a 6V nominal solar panel input.
- Battery Protection: Dedicated protection circuitry for over-charge, over-discharge, and short-circuit protection.
- Temperature Protection: The battery charging circuit must incorporate an NTC thermistor input to physically disable charging if the ambient temperature falls below 5°C or exceeds 55°C, preventing battery damage.
- Mechanical Power Switch: A physical switch (slide or toggle) to completely disconnect battery power from the system load (ESP32-C6, sensors, etc.). The solar charge controller must remain connected directly to the battery circuit ahead of the switch, ensuring the battery can continue solar harvesting even when the system load is manually switched off.
- Battery Monitoring: Circuitry to measure battery voltage via the ESP32-C6 ADC, switched through a MOSFET (high-side load switch) to eliminate parasitic leakage current when unread.
- Switchable 5V Sensor Power: Step-up (boost) converter generating 5V from the 3.7V battery to exclusively supply the external RS485 soil probe. To aggressively save energy, this 5V boost converter must be fully gateable (enable/disable) via a single ESP32-C6 GPIO pin. The ESP32-C6 will disable the boost converter completely during deep sleep to eliminate any standby current draw.

4. Sensor Interfaces, Expansion & UI

- Primary Sensor Port: Low-power RS485 transceiver with TVS protection on A/B lines for ESD and transient suppression.
- Status LEDs: Minimal diagnostic LEDs:
 - 1x Battery Charge Status LED (driven by the solar charge controller to indicate active charging/full).
 - 1x User-Programmable LED: Through-Hole (THT) component or dedicated 2-pin header for external panel mounting, controlled via ESP32-C6 GPIO.

Note: Diagnostic LEDs must be switchable or equipped with cut jumpers to prevent violating sleep current targets.
- Expansion I/O: Breakout header (2.54mm pitch) exposing all remaining unused ESP32-C6 GPIO pins, 3.3V, and GND.

5. PCB Layout & Form Factor

- Layer Count: Designer's discretion (2-layer or 4-layer) based on cost-efficiency and RF signal integrity requirements.
- RF Constraints: Adhere strictly to RF keep-out zones, proper trace impedance, and solid ground pours.
- Dimensions: Exact PCB dimensions and mechanical mounting constraints will be supplied prior to final routing. The initial layout should prioritize compactness.

6. Power Consumption Targets

- Deep Sleep Current: Total quiescent current of the assembled board (with sleeping ESP32-C6, LoRa module in sleep, and 5V boost/GPS subsystems gated off) must be $\leq 25 \mu\text{A}$.
- Active Efficiency: High-efficiency conversion across power regulators to maximize operational battery life.

7. Required Deliverables from Contractor

1. Native EDA design files (Altium, KiCad, or EasyEDA).
2. Complete Bill of Materials (BOM) with manufacturer part numbers and supplier references (e.g., LCSC or DigiKey).
3. Complete manufacturing package: Gerber files, Drill files, and Pick & Place (CPL) files.
4. Mechanical 3D Model: Complete .STEP assembly of the populated PCBA.
5. Board bring-up test firmware validating ESP32-C6 SPI communication with the SX1262, GPIO-controlled 5V power gating for energy savings, RS485 communication, and GPS power switching.